BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 2 Practical No: 2

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Subject Name : Deep Learning

Problem Statement:

Assignment 2. Design and implement a Multilayer Perceptron (MLP) for classification of the Iris or Wine dataset and evaluate its performance using accuracy and a confusion matrix.

Algorithm:

1. Start
2. Load the Wine Dataset
3. Split the dataset into input features (x) and output features (y)
4. Split the dataset into training and testing sets in the ratio 80:20
5. Create a MLP classifier with 2 layers (10 neurons each) and ReLU activation function.
6. Train the MLP classifier using training data
7. Test the model on the test dataset
8. Generate confusion matrix and classification report
9. Stop

Code and Output:

```
[ ]
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
[ ]
wine= load_wine()
x,y= wine.data, wine.target
x_train, x_test, y_train, y_test= train_test_split(x,y, test_size=0.2, random_state=20)
```

```
[ ]
scaler = StandardScaler()
X_train= scaler.fit_transform(x_train)
X_test= scaler.transform(x_test)
```

```
mlp= MLPClassifier(hidden_layer_sizes=(10,10), activation= 'relu', solver='adam', max_iter=1000, random_state=20)
mlp.fit(X_train,y_train)
y_pred=mlp.predict(X_test)
print("Accuracy:", accuracy_score(y_test,y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test,y_pred))
print("Classification Report:\n", classification_report(y_test,y_pred))
```

```
Accuracy: 0.9722222222222222
Confusion Matrix:
 [[10  0  0]
 [ 1 16  0]
 [ 0  0  9]]
Classification Report:
              precision    recall  f1-score   support

           0       0.91      1.00      0.95        10
           1       1.00      0.94      0.97        17
           2       1.00      1.00      1.00         9

    accuracy                           0.97        36
   macro avg       0.97      0.98      0.97        36
weighted avg       0.97      0.97      0.97        36
```